We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
Voraxor Herbicide

Use: crop protection product, herbicide

Manufacturer/supplier:
BASF Australia Limited (ABN 62 008 437 867)
Level 23, 40 City Road, Southbank
Victoria 3006, AUSTRALIA
Telephone: +61 3 8855-6600

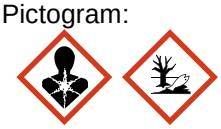
Emergency information:
BASF Emergency Advice Number: 1800 803 440 (24h) [within Australia]
BASF Emergency Advice Number: + 61 3 8855 6666 [outside Australia]

## 2. Hazard identification

Classification of the substance and mixture:
Reproductive toxicity: Cat.1B (fertility)
Reproductive toxicity: Cat.2 (unborn child)
Specific target organ toxicity — repeated exposure: Cat.2
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:



Signal Word:

Danger

Hazard Statement:

| | |
|---|---|
| H373 | May cause damage to organs through prolonged or repeated exposure. |
| H360 | May damage fertility. Suspected of damaging the unborn child. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P260 | Do not breathe mist or vapour. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |

Precautionary Statements (Response):

| | |
|---|---|
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P391 | Collect spillage. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

May produce an allergic reaction. Contains:
1,2-benzisothiazol-3(2H)-one, mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, herbicide

**Hazardous ingredients**

saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide

| | |
|---|---|
| Content (W/W): 21.51 % | Repr.: Cat. 2 (unborn child) |
| CAS Number: 372137-35-4 | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |

trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione

| | |
|---|---|
| Content (W/W): 10.76 % | Repr.: Cat. 1B (fertility) |
| CAS Number: 1258836-72-4 | STOT RE: Cat. 2 |
| | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 1000 |
| | M-factor chronic: 100 |

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen fluoride, hydrogen bromide, hydrogen chloride, nitrogen oxides, silicon oxides, sulfur oxides, halogenated compounds, cyanides
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray.Use personal protective clothing.Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into the subsoil/soil.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

## 8. Exposure controls and personal protection

Components with occupational exposure limits

propane-1,2-diol, 57-55-6;
TWA value 474 mg/m3 ; 150 ppm (AU NOEL), Total vapour and particulates
TWA value 10 mg/m3 (AU NOEL), Particulate
TWA value 474 mg/m3 ; 150 ppm (OEL (AU)), Total vapour and particulates
TWA value 10 mg/m3 (OEL (AU)), Particulate

Saflufenacil, 372137-35-4;
TWA value 0.824 mg/m3 (BASF recomm. occupational exposure limit)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid |
| Colour: | off-white |
| Odour: | faint, sweetish |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | approx. 5 - 7<br>(100 %(m), 23 °C) |
| Freezing point: | approx. -5 °C |

Boiling point: > 110 °C

Flash point:
Non-flammable.
Evaporation rate:
not applicable

Flammability (solid/gas): not applicable
Lower explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Upper explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Ignition temperature: 460 °C

Thermal decomposition: > 190 °C , 620 kJ/kg (DSC (OECD 113))

Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.
SADT: > 75 °C
Explosion hazard: not explosive
Fire promoting properties: not fire-propagating

Vapour pressure: approx. 23 hPa
(20 °C)
Information applies to the solvent.

Density: approx. 1.16 g/cm3
(20 °C)
Relative vapour density (air):
not applicable

Solubility in water: dispersible
Partitioning coefficient n-octanol/water (log Pow):
The statements are based on the properties of the individual components.
Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Partitioning coefficient n-octanol/water (log Pow): 2.6
(20 °C; pH value: 1.7)
---

Viscosity, dynamic: approx. 58 mPa.s
(20 °C, 200 1/s)

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

## 10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

| | |
|---|---|
| Thermal decomposition: | > 190 °C, 620 kJ/kg (DSC (OECD 113)) |
| Thermal decomposition: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

**Acute oral toxicity**

Experimental/calculated data:
LD50rat (oral): > 2,000 mg/kg
No mortality was observed.

**Acute inhalation toxicity**

LC50 rat (by inhalation): > 3.4 mg/l 4 h
Highest concentration technically achievable. An aerosol was tested.

**Acute dermal toxicity**

LD50 rat (dermal): > 5,000 mg/kg
No mortality was observed.

**Assessment of acute toxicity**

Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

## Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: non-irritant

## Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
Buehler test guinea pig: Non-sensitizing.

## Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bronopol
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was mutagenic in various cell culture test systems; however, these results could not be confirmed in tests with mammals.
---

## Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Assessment of carcinogenicity:
In long-term studies in rats the substance induced thyroid tumors. The effect is caused by an animal specific mechanism that has no human counter part. Due to the species specific mode of action, the effects are not expected to occur in humans.
---

## Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Assessment of reproduction toxicity:
Causes impairment of fertility in laboratory animals.
---

### Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.
---

### Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Assessment of repeated dose toxicity:
Repeated oral exposure may affect certain organs.

Information on: bronopol
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Information on: mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. Based on available data, the classification criteria are not met.
---

### Aspiration hazard

not applicable

### Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Toxicity to fish:
LC50 (96 h) > 1.76 mg/l, Oncorhynchus mykiss

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Toxicity to fish:
LC50 (96 h) > 96.8 mg/l, Pimephales promelas (OECD Guideline 203, static)
---

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Aquatic invertebrates:
EC50 (48 h) > 1.95 mg/l, Daphnia magna

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Aquatic invertebrates:
EC50 (96 h) 8.0 mg/l, Americamysis bahia (static)
---

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Aquatic plants:
EC50 (72 h) 0.000821 mg/l (growth rate), Pseudokirchneriella subcapitata

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Aquatic plants:
EC50 (96 h) 0.113 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
---

### Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Assessment transport between environmental compartments:
Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.
---

## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
---

## Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: saflufenacil (ISO); N´-{2-chloro-4fluoro-5-[1,2,3,6-tetrahydro-3-methyl-2,6-dioxo-4-(trifluoromethyl)pyrimedin-1-yl]benzoyl}-N-isopropyl-N-methylsulfamide
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.
---

Information on: trifludimoxazin (ISO); 1,5-dimethyl-6-thioxo-3-[2,2,7-trifluoro-3-oxo-4-(prop-2-yn-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-1,3,5-triazinane-2,4-dione
Bioaccumulation potential:
Bioconcentration factor: 81.6
---

## Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (SAFLUFENACIL, TRIFLUDIMOXAZINE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Further information**
Hazchem Code:3Z
IERG Number:47

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (SAFLUFENACIL, TRIFLUDIMOXAZINE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (SAFLUFENACIL, TRIFLUDIMOXAZINE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

**Further information**
Environmentally Hazardous Substances meeting the description of UN 3077 or UN 3082 are not subjected to the Australian Dangerous Goods Code when transported by road or rail in packagings not exceeding 500 kg(L) or IBCs.

**Further information**
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

**Other regulations**

To avoid risks to man and the environment, comply with the instructions for use.

Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP): Schedule 5

APVMA Approval No.: 86452

**Registration status:**

AICIS, AU

Contains non-registered, non-listed substance., Individual registration may be required., Please contact your BASF representative.

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.